Yi Yuan

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Male | Ph.D. Candidate in Physics, Tsinghua University

Education

Tsinghua University Ph.D. in Physics 2023.08 – 2028.06 (Expected)
Department of Physics Ranked **1st** in Modern Physics in the Ph.D. Qualifying Exam Principles and Architecture of Quantum Computers (A+) Beijing

B.S. in Physics, Fundamental Science Experimental Class (Mathematics and Physics), Tsinghua University 2019.08 – 2023.06
Department of Physics GPA: 3.89/4.0 (Rank: 18/116) Outstanding Graduate Nominee, Ye Qisun Award Beijing

Technical Skills

- **Programming Languages:** Python (primary language for daily research and development)
- **AI/ML:** Familiar with Transformer architectures; experienced with JAX and PyTorch; hands-on experience in CNN modeling
- **Data and Computing:** NumPy vectorized computing, multiprocessing, data-driven modeling, and backtesting system development
- **Tools:** Git, Linux servers **Language:** English – proficient in reading and writing scientific literature

Publications

Yuanchen Zhao*, **Yi Yuan***, Zhengyi Han, and Dong E. Liu. "Single-Shot State Preparation Threshold: Rigorous Theorem and Statistical Mechanical Mapping." *arXiv preprint arXiv:2609.39965*, 2026. arXiv *Equal contribution.
Gave sufficient conditions for single-shot state preparation in QEC codes and mapped the problem to statistical mechanical models; mainly responsible for the statistical mechanical mapping and numerical calculations.

Yi Yuan, Yuanchen Zhao, and Dong E. Liu. "Zeno-Enhanced Probabilistic Error Cancellation with Quantum Error Detection Codes." *Quantum Science and Technology*, accepted, 2026. arXiv
Developed a perturbative QEDC+PEC protocol for near-term noisy quantum devices, using post-selection to reshape the effective noise channel and PEC to cancel residual logical errors.

Research and Projects

Research in Quantum Computing and Quantum Information 2022.09 – Present
Department of Physics, Tsinghua University Python, JAX, Tensor Network

- **Zeno-enhanced QEDC+PEC protocol** (2024 – 2026, accepted in *Quantum Science and Technology*): Combined quantum error-detecting codes with probabilistic error cancellation for near-term noisy quantum devices; in 200-qubit Iceberg-code logical GHZ preparation, reduced sampling overhead by **3–4 orders of magnitude** relative to standard PEC while maintaining $F \simeq 0.956$
- **Classical Shadow error mitigation** (2024): Used tensor network methods to compute the error-mitigation complexity of classical shadows on noisy quantum circuits with $\log(n)$ depth, and analyzed its scaling with system size
- **Undergraduate thesis: complexity of quantum annealing** (2022.09 – 2023.06): Based on the imaginary-time spacetime instanton action, explained why quantum annealing and classical QMC have matching complexity in ground-state energy search for double-well potentials, and identified the possibility of multi-path quantum speedup

Selected Projects

- **SEVI particle scattering analysis**: Built a PyTorch-based CNN to fit scattering-amplitude parameters, reaching the accuracy of analytical methods without injecting prior knowledge
- **LLM paper summarization tool** (2025): Automatically fetched daily arXiv papers, generated summaries through the Kimi API, and completed local deployment

Honors and Leadership

Honors: 2021 Tsinghua Alumni – Zhang Mingwei Scholarship Multiple Department of Physics scholarships for academic excellence, innovation, and overall achievement in 2020–2021
Leadership: 2023.08–2025.01 Student Counselor, Department of Physics, Tsinghua University 2024–2025 Outstanding Student Leader